

Michał Weiss

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Education

- 2025–2029 **Doctoral studies** (in progress) — *University of Warsaw*
Interdisciplinary Doctoral School (Międzydziedzinowa Szkoła Doktorska); disciplines: mathematics and psychology. Home unit: Faculty of Psychology, Human Interactivity and Language Lab (HILL).
Working topic: interpersonal coordination and modes of agency in Capoeira Angola — movement dynamics paired with first- and second-person accounts of experience.
Supervisor: prof. Joanna Rączaszek-Leonardi (UW). Foreign co-supervisor: prof. Michael J. Richardson (Macquarie University).
- 2023–2025 **Master of Arts in Cognitive Science** — *University of Warsaw, Faculty of Psychology*
Graduated *very good with distinction*.
Thesis: *Quantifying Movement Coordination in Human–Robot Interaction*. Methods: Recurrence Quantification Analysis (RQA), nonlinear dynamics.
- 2017–2021 **Bachelor of Science in Computer Science (Engineering)** — *Warsaw University of Technology, Faculty of Electrical Engineering*
Specialization: Software Engineering.
Thesis: *Route Optimization using the Ant Colony Algorithm*.

Research interests

- movement coordination, embodied and distributed cognition
- contemplative studies; contemplative practice as relational and community-driven
- multimodal interaction (gaze, speech, movement)
- quantitative analysis of behaviour (time series, recurrence methods)
- mixed methods: dynamical measures paired with micro-phenomenological interviewing
- embodied mathematics learning and educational technology

Grants and awards

- 2026–2028 **European Varela Award** — *Mind & Life Europe*
From Center to Circle: Capoeira Angola as a Community-Driven, Relational Contemplative Practice — A Mixed-Methods Account. Principal Investigator. 24 months (X.2026–IX.2028). Fieldwork in Salvador, Bahia: IMU and markerless motion capture combined with Immediate Experience Reports and video-cued micro-phenomenological interviews.
- 2023 **2nd place, international hackathon** — *VR Summit 2023, Bochum, Germany*

Publications

- in press Abrahamson, D., Tancredi, S., Xiao, A., Li, X., **Weiss, M.**, Potega vel Żabik, K., & Dimmel, J. K. (2026). It's Our Move: Mathematical Perception Emerges for Coordinating Joint Action. *Canadian Journal of Experimental Psychology*.
- preprint **Weiss, M.**, Araújo, E., Silva, P. A., Bermudez i Badia, S., Pilacinski, A., & Zubek, J. (2024). Quantifying Movement Coordination in Human–Robot Interaction.

Open-source tools.

Miao, G. Q., Pouw, W., & **Weiss, M.** *DIMS Dashboard* — an open tool for visualizing multimodal social interaction data: time series, video, transcripts, and optional analyses (RQA, cross-wavelet, ELAN

annotations). Deployed for several research corpora, including the OЯTHO study and a Karnataka music corpus.

Research experience

- 2025–now **Research internship** — *Copernicus Science Centre, Warsaw*
Embodied mathematics learning at the OЯTHO exhibit: how children come to perceive a Cartesian coordinate system while coordinating joint action. Corpus of over 1,200 players; analysis of task-order effects. Basis of the *Canadian Journal of Experimental Psychology* paper above.
- 2025–now **Doctoral research embedded in NCN Opus 27 “CoMotion”** — *HILL, Faculty of Psychology, UW*
Movement coordination: between interpersonal connection and individual autonomy (PI: dr Julian Zubek). Methodological and infrastructural affiliation; coupled ↔ decoupled transitions as markers of interaction quality.
- 2025 **Triadic coordination study** — *HILL, UW, with dr Julian Zubek*
Interpersonal coordination in father–child–mother triads, analysed with Cross-Recurrence Quantification Analysis.
- 2023–2024 **Research internship (50 h)** — *NCN Sonatina project 2022/44/C/HS6/00068*
Study of the sense of temporal continuity between episodes of altered perceptual experience. Data preprocessing and analysis scripts for ECG (Biopac Bionomadix) and eye-tracking (HTC Vive Pro Eye).

Teaching — university

- 2025/2026 **Cognitive Processes Modeling II** — *University of Warsaw*
Lead lab instructor on the Cognitive Science MA programme (60 h, 6 ECTS). Modelling categorisation, memory, decision making and motor coordination — hands-on, in code.
- 2025/2026 **Learning, adaptation and uncertainty: machine learning for cognitive modeling** — *University of Warsaw*
Co-taught, including part of the lecture series. Reinforcement learning, swarm algorithms, Kalman filters.

Teaching — schools and tutoring

- 2020–now **Teacher of Mathematics and Computer Science** — *Classical School (formerly Akademia Tutoring College), Warsaw*
iGCSE and A-level preparation for admission to top universities; taught in English.
- 2025/2026 **Course tutor in Mathematics** — *Symposio*
Four groups of 8–20 students, preparing for the year-8 national examination and the extended-level *matura*.
- 2021–2022 **Teacher of Mathematics and Computer Science** — *Liceum w Chmurze, Warsaw*
An on-site high school in the Słodowiec district, despite the name.
- 2021–2022 **Teacher of Mathematics and Computer Science** — *L.O. im. Zuzanny Ginczanki, Warsaw*
Minority education programme.
- Otwarte Centrum Edukacyjne** — *home-education centre, Ursynów, Warsaw*
Preparing home-schooled students for the Computer Science examination.
- 8+ years **Private tutoring at every level** — primary school to university. Mathematics for the year-8 examination and the *matura*; statistics and data analysis for BA and MA students in psychology, cognitive science and the social sciences (R, Python, SPSS, JASP; APA reporting); programming and computer science.

Supervised student projects — graph-based analysis of a corpus of roughly 60,000 perfume compositions (SQL, pandas, NetworkX); a Flask delivery-planning web application with a hand-written SQL schema and nearest-neighbour route heuristics.

Workshops and mentoring — youth workshops in programming, logic and modelling; mentor in programming contests and educational projects.

Conferences, summer schools and academic service

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| 2026 | ESRI 2026 — European Summer Research Institute and the Mind & Life Europe Retreat, Scharnstein, Austria. Poster: <i>From Center to Circle</i> . |
| 2025 | Unboxing Multimodality — summer school, University of Amsterdam. |
| 2024 | 13th Peripatetic Conference on Cognitive Systems Modeling , Zakopane. Presentation: <i>Quantifying Movement Coordination in Human–Robot Interaction</i> . |
| 2023 | 4th Summer School on Social Human–Robot Interaction , Poland — co-organizer and participant, poster presentation. |
| 2023 | 12th Peripatetic Conference , Zakopane — organized the workshop “ <i>What is the shape of doubt in the mind of a cognitive scientist?</i> ” |
| 2023 | NLM 2023 , Leuphana University — summer school on RQA, fractal analysis and convergent cross-mapping. |
| 2023 | Cognitive Futures in the Arts and Humanities , University of Warsaw — volunteer. |
| 2022 | 11th Peripatetic Conference on Cognitive Systems Modeling , Zakopane. Poster: <i>Fractals — a formula for creativity</i> . |

Skills

- programming: Python, R, Java, Julia, SQL; Flask, Plotly Dash
- time-series and dynamical systems analysis (RQA, CRQA, fractal analysis, convergent cross-mapping)
- digital signal processing: motion capture, eye-tracking, ECG
- motion capture: IMU (Perception Neuron), markerless pose estimation (YOLO, RTMPose, Pose2Sim), OpenSim; ELAN annotation
- statistics for the social sciences: regression, ANOVA, mixed models, effect sizes, APA reporting (R, Python, SPSS, JASP)
- interactive data visualization (Plotly Dash, matplotlib, ggplot2, seaborn)
- Git and GitHub Actions, GNU/Linux, L^AT_EX

Languages

Polish (native) · English (C1/C2) · Portuguese (basic, in progress)

Other interests

Beyond my academic work, I am interested in the dialogue between psychology and religion. I explore different perspectives on the human self, both outward and inward, drawing inspiration from Advaita Vedanta, Taoist philosophy, and the tradition of psychoanalysis. This runs alongside my own contemplative practice (Qi Gong, Jnana Yoga) and my practice of Capoeira Angola, which is also the field of my doctoral work.